DETERMINING OPEN CIRCUIT LOCATION (MS-CAN) [SKYACTIV-D 1.5]

id100218000500

Caution

- **Perform the following malfunction diagnosis only when it is diagnosed with a open circuit by CONTROLLER AREA NETWORK (CAN) MALFUNCTION DIAGNOSIS FLOW. (See CONTROLLER AREA NETWORK (CAN) MALFUNCTION DIAGNOSIS FLOW [SKYACTIV-D 1.5].)**
- **If the malfunctioning part is detected in the communication line, before disconnecting the related connector for inspection, press the connector in the connection direction to verify that there is no looseness or disconnection.**
- **When disconnecting the connector, verify that there is no damage, deformation, or corrosion of the connector terminals.**

1. Verify DTCs of the modules related to the CAN system.
2. Apply the communication error DTC and the failed module to DTC output pattern and malfunctioning location, and select the possible cause for the diagnostic result and the reference for the inspection item. (See DTC Output Pattern And Malfunctioning Location.)

Note

- The open circuit location can be determined by the DTC indicated in the DTC output pattern and malfunctioning location chart. DTCs not listed in the chart are not used for the determination of the open circuit location. (See DTC Output Pattern And Malfunctioning Location.)

3. Inspect the possible cause and inspection item of the applicable malfunctioning part.
4. After repairs, return to CONTROLLER AREA NETWORK (CAN) MALFUNCTION DIAGNOSIS FLOW, and verify that the repairs have been completed. (See CONTROLLER AREA NETWORK (CAN) MALFUNCTION DIAGNOSIS FLOW [SKYACTIV-D 1.5].)

DTC Output Pattern And Malfunctioning Location

Cross (×): Communication error-related DTC and failed module

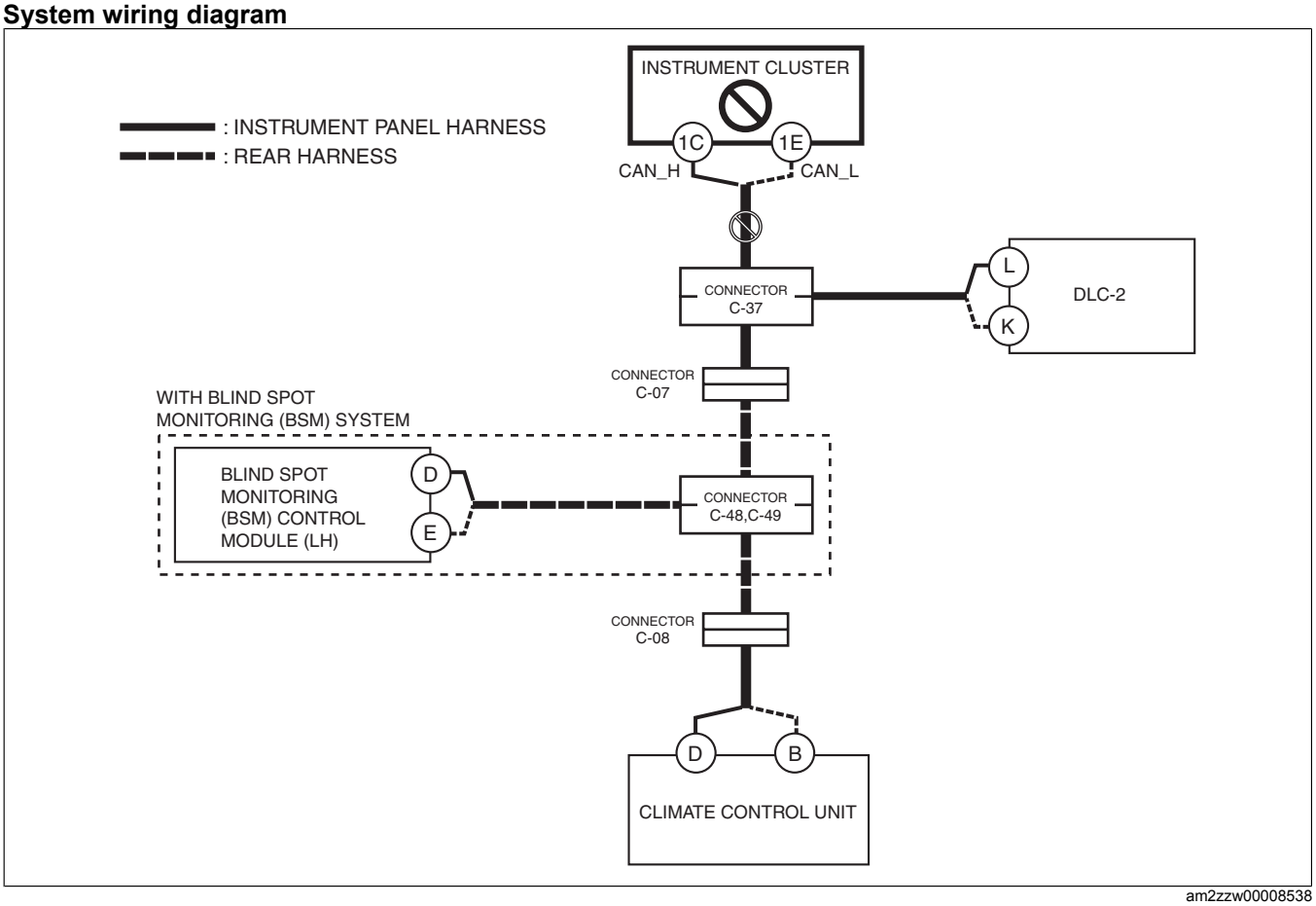
M-MDS display		DTC output pattern and malfunctioning location			
DTC output module	**DTC**				
IC (Instrument cluster)	U0232:00		×	×	
BSM*1 (Blind spot monitoring (BSM) control module (LH))	U0100:00	×			
	U0121:00	×			
	U0155:00	×			
	U0214:00	×			
EATC*2 (Climate control unit)	U0155:00	×			
M-MDS display module		**[Fail] display pattern**			
IC		×			
BSM*1			×	×	
EATC*2			×		×
Diagnostic result					
Possible cause and inspection item		A	B	C	D

*1 : With blind spot monitoring (BSM) system
*2 : With full-auto air conditioner

A
Possible cause

- Connector terminal disconnection, poor contact, damage, deformation, corrosion
- Instrument cluster power supply voltage or body ground malfunction
- Open circuit in wiring harness between instrument cluster and connector C-37
- Connector C-37 malfunction
- Instrument cluster malfunction

System wiring diagram



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Inspection item

- Instrument cluster power supply voltage-related wiring harness and fuse
- Instrument cluster body ground related wiring harness
- Instrument cluster connector
- Connector C-37
- Wiring harness between instrument cluster terminal 1C and connector C-37
- Wiring harness between instrument cluster terminal 1E and connector C-37
- Instrument cluster

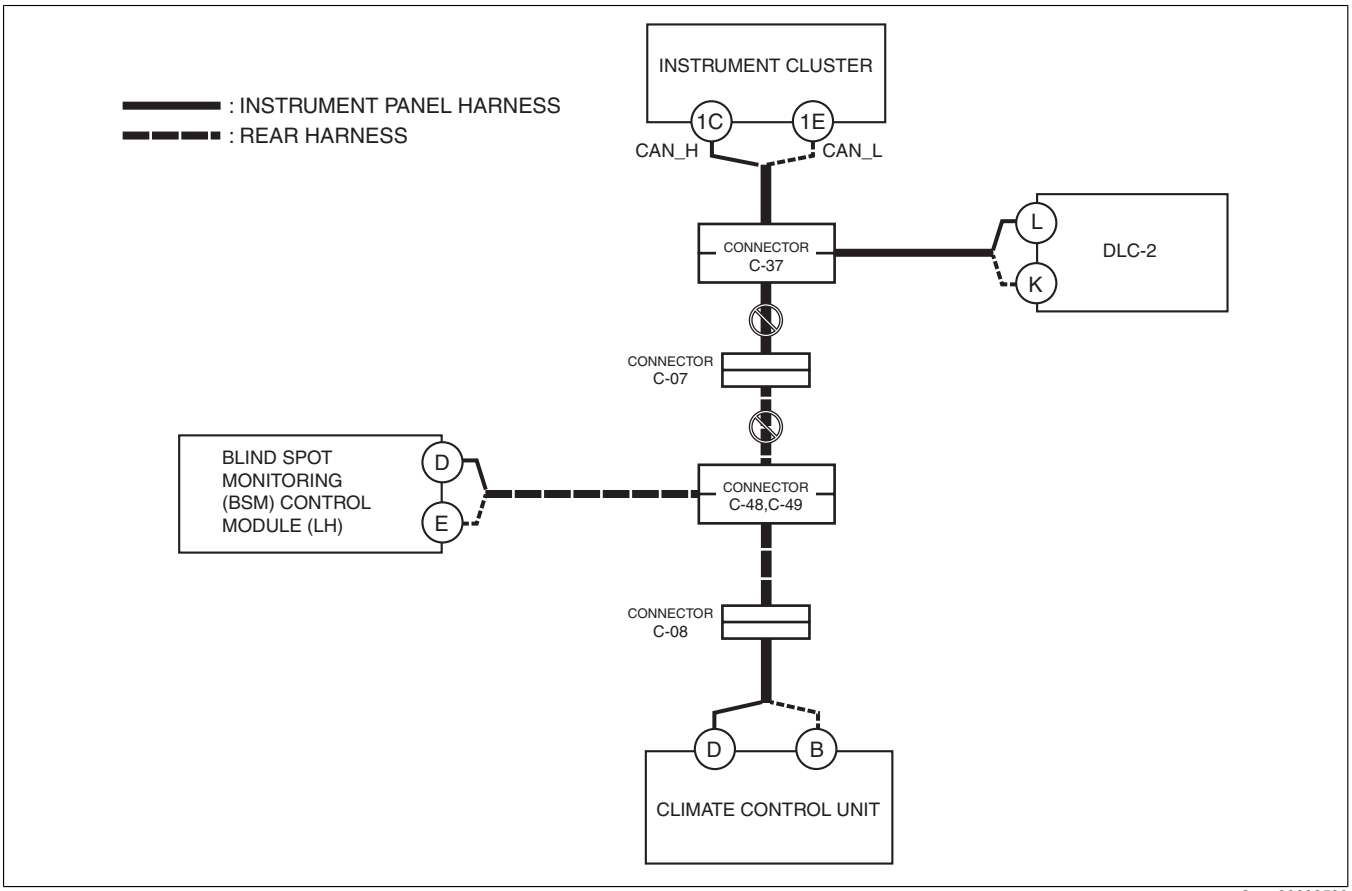
B

With blind spot monitoring (BSM) system

Possible cause

- Connector terminal disconnection, poor contact, damage, deformation, corrosion
- Open circuit in wiring harness between connector C-37 and connector C-07
- Open circuit in wiring harness between connector C-07 and connectors C-48 and C-49
- Connector C-37 malfunction
- Connector C-07 malfunction
- Connectors C-48 and C-49 malfunction

System wiring diagram



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Inspection item

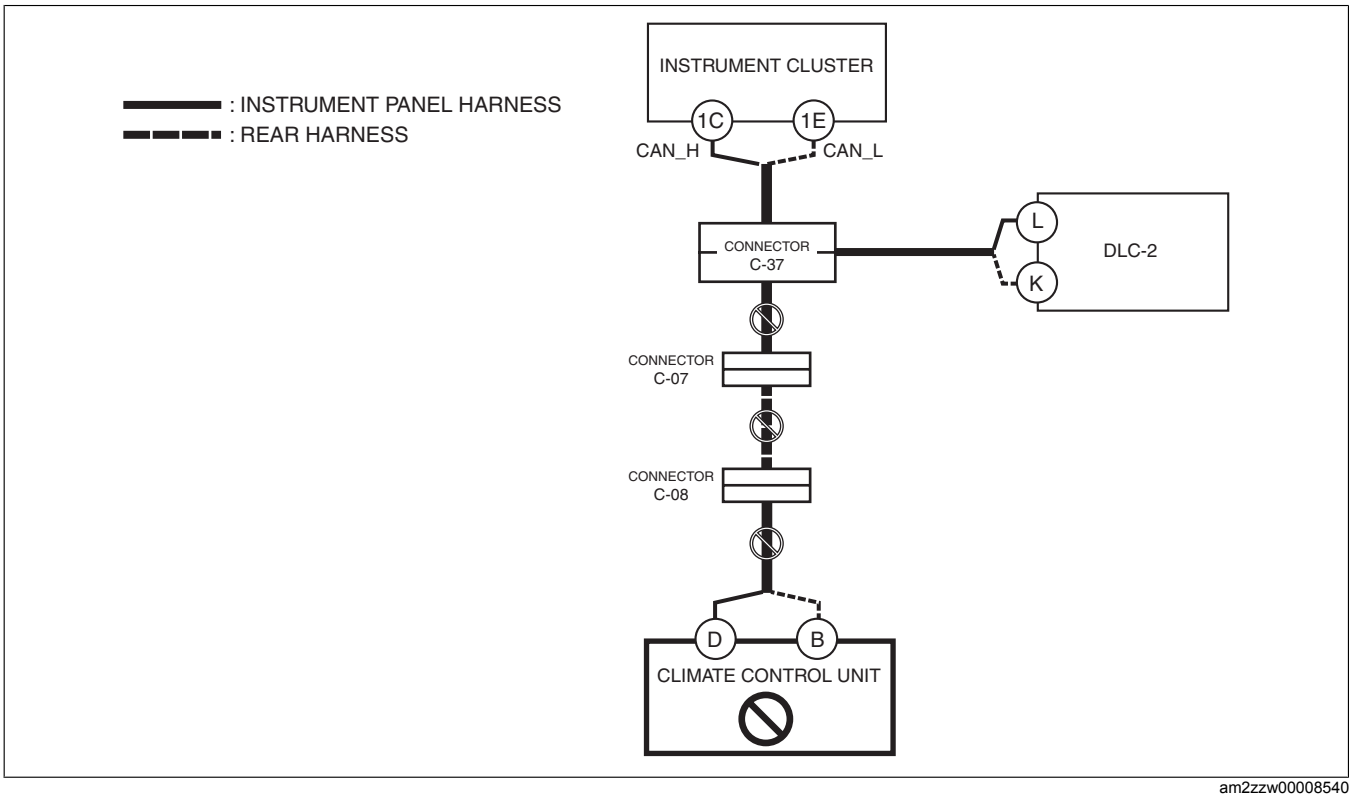
- Connector C-37
- Connector C-07
- Connectors C-48 and C-49
- Wiring harness between connector C-37 and connector C-07
- Wiring harness between connector C-07 and connectors C-48 and C-49

Without blind spot monitoring (BSM) system

Possible cause

- Connector terminal disconnection, poor contact, damage, deformation, corrosion
- Climate control unit power supply voltage or body ground malfunction
- Open circuit in wiring harness between connector C-37 and connector C-07
- Open circuit in wiring harness between connector C-07 and connector C-08
- Open circuit in wiring harness between climate control unit and connector C-08
- Connector C-37 malfunction
- Connector C-07 malfunction
- Connector C-08 malfunction
- Climate control unit malfunction

System wiring diagram



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Inspection item

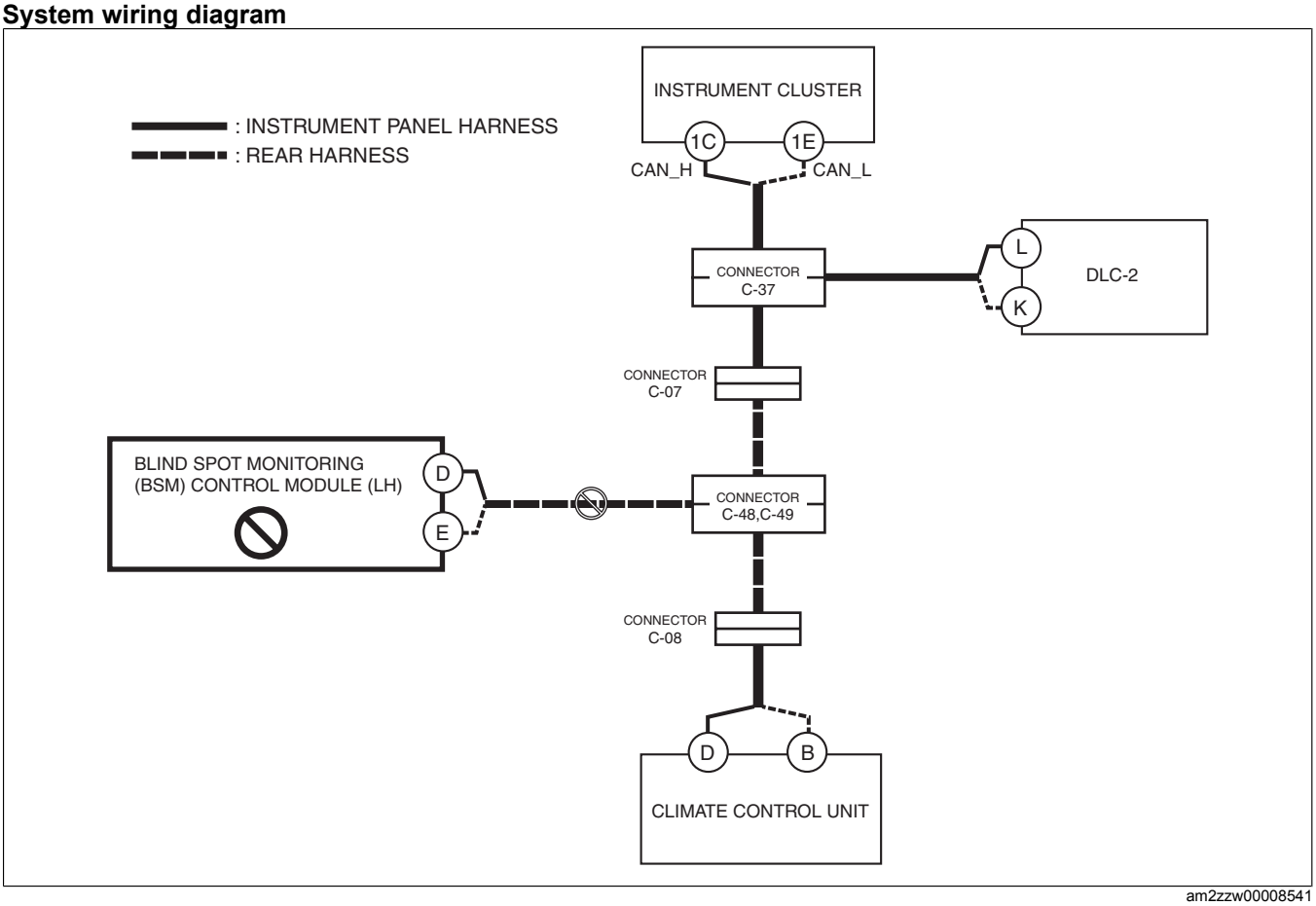
- Climate control unit power supply voltage-related wiring harness and fuse
- Climate control unit body ground related wiring harness
- Climate control unit connector
- Connector C-37
- Connector C-07
- Connector C-08
- Wiring harness between connector C-37 and connector C-07
- Wiring harness between connector C-07 and connector C-08
- Wiring harness between climate control unit terminal D and connector C-08
- Wiring harness between climate control unit terminal B and connector C-08
- Climate control unit

C

Possible cause

- Connector terminal disconnection, poor contact, damage, deformation, corrosion
- Blind spot monitoring (BSM) control module (LH) power supply voltage or body ground malfunction
- Open circuit in wiring harness between blind spot monitoring (BSM) control module (LH) and connectors C-48 and C-49
- Connectors C-48 and C-49 malfunction
- Blind spot monitoring (BSM) control module (LH) malfunction

System wiring diagram



am2zzw00008541

Inspection item

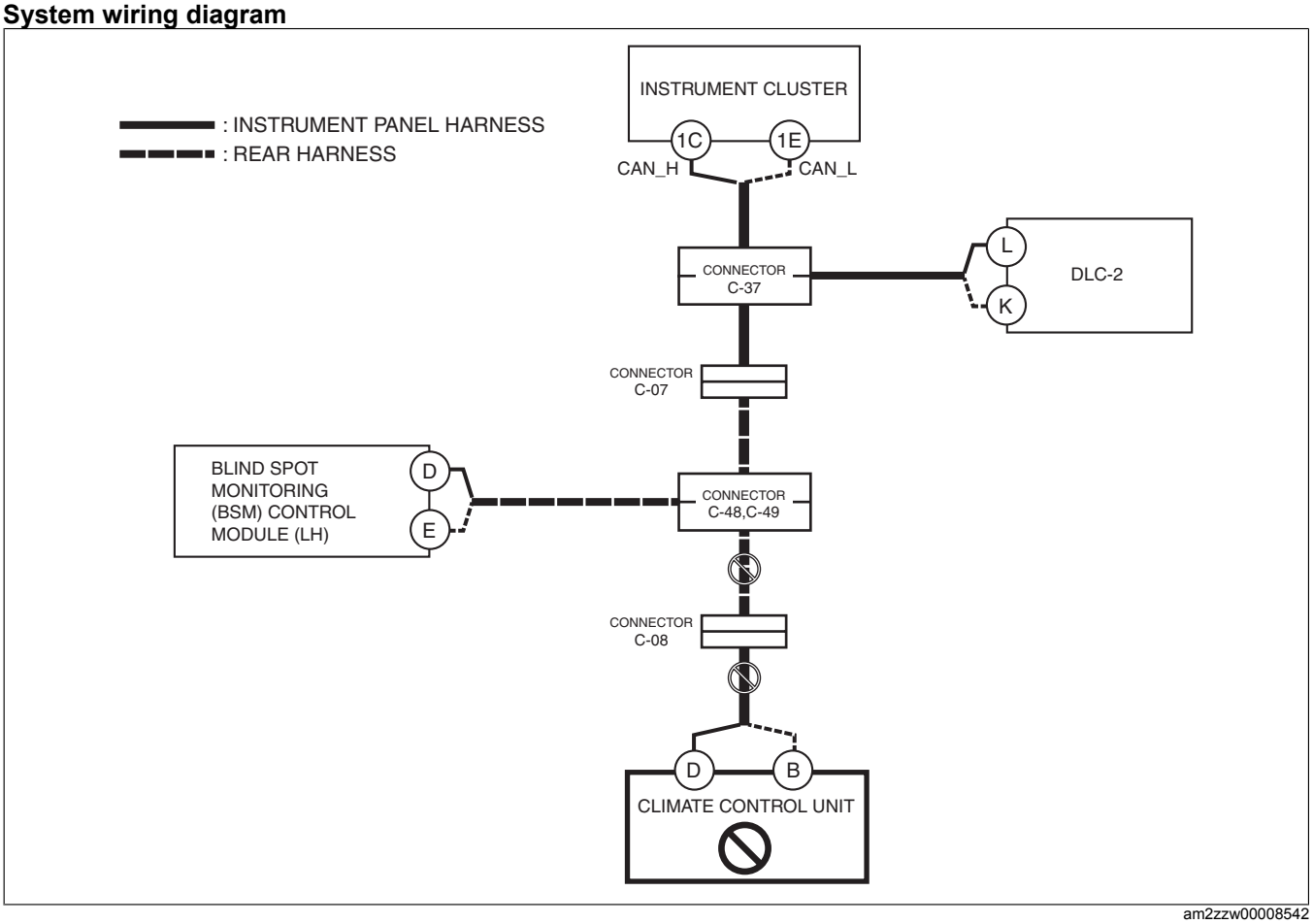
- Blind spot monitoring (BSM) control module (LH) power supply voltage-related wiring harness and fuse
- Blind spot monitoring (BSM) control module (LH) body ground related wiring harness
- Blind spot monitoring (BSM) control module (LH) connector
- Connectors C-48 and C-49
- Wiring harness between blind spot monitoring (BSM) control module (LH) terminal D and connector C-48
- Wiring harness between blind spot monitoring (BSM) control module (LH) terminal E and connector C-49
- Blind spot monitoring (BSM) control module (LH)

D

Possible cause

- Connector terminal disconnection, poor contact, damage, deformation, corrosion
- Climate control unit power supply voltage or body ground malfunction
- Open circuit in wiring harness between connectors C-48 and C-49 and connector C-08
- Open circuit in wiring harness between connector C-08 and climate control unit
- Connectors C-48 and C-49 malfunction
- Connector C-08 malfunction
- Climate control unit malfunction

System wiring diagram



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Inspection item

- Climate control unit power supply voltage-related wiring harness and fuse
- Climate control unit body ground related wiring harness
- Climate control unit connector
- Connector C-08
- Connectors C-48 and C-49
- Wiring harness between connectors C-48 and C-49 and connector C-08
- Wiring harness between climate control unit terminal D and connector C-08
- Wiring harness between climate control unit terminal B and connector C-08
- Climate control unit